

Marko_Mitrovic_CV

Marko Mitrović

Software Engineer | Backend & Legacy Modernization Specialist

Location: Novi Sad, Serbia

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Technical Focus: PHP, Java, Spring Boot, React, Flutter, Oracle SQL, MariaDB (MySQL), SSH Deployment

“Pragmatic Software Engineer specializing in the modernization of large-scale retail ecosystems and enterprise-grade systems. Expert in executing multi-track architectural migrations, transitioning complex legacy environments to unified Java and modern PHP platforms with a zero-downtime objective. Proven track record in maintaining high-traffic infrastructure where system availability and data integrity are non-negotiable. Skilled in architecting robust integration layers (SAP, PowerBI, E-commerce APIs) and aligning technical infrastructure with rigorous regulatory and audit standards.”

TECHNICAL SKILLS

- **Programming Languages:** Java, PHP, SQL (PL/SQL), Dart, JavaScript.
- **Frameworks & Ecosystems:** Spring Boot (JPA, Security), Flutter, React, Node.js.
- **Databases & Data Architecture:**
 - **Relational:** Oracle (Performance Tuning & Data Integrity), MariaDB, MySQL.
 - **NoSQL & Cache:** Redis, MongoDB, Cassandra, Neo4j (Graph Database).
- **Enterprise Integrations:** SAP ERP (RESTful API), Enterprise Data Pipelines, PowerBI Analytics, Last-mile delivery orchestration (Glovo, Wolt).
- **Cloud & DevOps:** Microsoft Azure, Docker, Nginx, Apache, CI/CD workflows, Linux (Server Administration).
- **Systems Engineering:** Distributed Locking, Audit Logging Systems, Regulatory Compliance, High-Availability Design, System Hardening.

PROFESSIONAL EXPERIENCE

Software Engineer | Univerexport DOO, Novi Sad

December 2023 — August 2026

Key Focus: Leading multi-track architectural migration, enterprise-scale integrations, and system compliance.

- **Legacy Modernization & Dual-Stack Evolution (PHP & Java):**
 - **Multi-Stage Ecosystem Migration:** Executing a 3-track strategic transition:
 - **Enterprise PHP Modernization:** Successfully migrated high-traffic modules to the latest stable PHP environments, ensuring modern security standards and significantly improved execution speed.
 - **PHP to Java:** Transitioning core business logic to a unified Spring Boot platform for enhanced scalability.
 - **Mission-Critical System Continuity:** Ensuring high availability and peak performance of core business modules during architectural transitions, focused on maintaining system resilience and uninterrupted service delivery during infrastructure scaling.
 - **Distributed Locking Service:** Engineered a custom cross-platform locking mechanism to resolve complex race conditions. This solution synchronized state between mobile, web, and backend environments, ensuring data consistency and preventing concurrent write conflicts across the entire ecosystem.
- **Enterprise Integration & Automation:**
 - **Financial Systems Interoperability:** Architected and maintained secure integration layers with major factoring platforms, enabling automated financial processing and reliable data exchange while strictly adhering to high-level security and audit standards.
 - **Enterprise Data Automation Engine:** Architected an automated pipeline for extracting and transforming large-scale datasets into optimized structures for downstream consumption. Implemented automated schema validation and multi-layered integrity checks to ensure data precision and secure integration with enterprise services.

- **Delivery Platforms:** Engineered robust integration layers for real-time SKU, pricing, and inventory synchronization between core databases and delivery partner platforms. Implemented high-performance batch processes to ensure data integrity and near real-time consistency across the distribution network.
- **Internal Tooling & Backend Standardization:**
 - **Developer Efficiency Suite:** Designed and implemented modular tools to streamline transition processes to modern backend architectures:
 - **Architectural Transition Tools:** Developed modular components to facilitate the migration toward modern backend standards.
 - **Meta-data Driven Reporting:** Implemented a high-performance reporting module leveraging **Spring Reflection & Custom Annotations** to decouple data formatting from business logic.
 - **API Contract Standardization:** Established a unified response architecture for the Spring Boot ecosystem, ensuring predictable data delivery and consistent interface contracts.
 - **Fluent Query Specifications:** Engineered a reusable component for **strongly-typed criteria filtering**, optimizing dynamic search performance while enforcing strict type-safety.
- **Enterprise System Interoperability:** Engineered **RESTful API** bridges with **Enterprise Resource Planning (ERP)** systems and optimized data pipelines for **Business Intelligence (BI)** analytics platforms, enabling seamless data orchestration and automated reporting.
- **Compliance & Security Engineering:**
 - **Governance & Standards** Aligning system architecture with industry regulations to ensure continuous technical compliance.
 - **Architectural Observability:** Designed and implemented advanced **telemetry and logging frameworks** to ensure system-wide transparency and operational accountability..
 - **Identity & Access Management (IAM):** Enforced **Principle of Least Privilege (PoLP)** through granular access models and schema hardening.
- **Data Orchestration & Infrastructure:**
 - **Distributed Data Synchronization:** Engineered a robust ecosystem of synchronization workers to maintain real-time consistency across **heterogeneous database environments**.
 - **Performance Engineering:** Optimized cross-instance data flow and query execution to minimize latency during high-frequency transfers in **distributed architectures**.
- **Mobile Ecosystem & Digital Transformation:**
 - **Enterprise E-commerce Platform:** Lead Architect for the official cross-platform mobile application (iOS & Android).
 - **Strategic Collaboration:** Bridging the gap between technical feasibility and business requirements to define UI/UX standards and digital brand identity.
 - **Performance & Integration:** Engineering high-performance communication layers between mobile frontends and modernized backends for real-time **order orchestration** and personalized user experiences.
 - **Industrial Solutions:** Developed and deployed specialized applications for industrial **PDA hardware**, optimizing warehouse logistics, scanning workflows, and real-time inventory tracking.

PROFESSIONAL INTERNSHIPS

DevOps Intern | Levi9, Novi Sad

May 2023 — June 2023

- **Infrastructure Learning Path:** Gained hands-on experience with **Microsoft Azure** services, focusing on resource provisioning and cloud environment management.
- **Infrastructure as Code (IaC):** Assisted in developing and maintaining cloud resources using **Terraform**, learning how to apply IaC principles for scalable environments.
- **Deployment Automation:** Participated in building and testing automated pipelines in **Azure DevOps** to understand the full CI/CD lifecycle.
- **Scripting & Support:** Utilized **PowerShell** for basic automation of routine administrative tasks and resource monitoring under senior guidance.
- **Agile Integration:** Worked within a **Scrum** team, gaining exposure to sprint cycles, daily stand-ups, and collaborative infrastructure development.

August 2022 - September 2022

- **Legacy-to-Modern Migration:** Spearheaded the transition of an internal project from Firebase to a **Laravel** backend, significantly improving system stability and data security.
- **Full-Stack Development:** Modernized the frontend by implementing **Tailwind CSS**, enhancing the user experience (UX) for city landmark management tools.
- **Database Management:** Managed SQL migrations, ensuring data integrity during the architecture overhaul.

EDUCATION & LANGUAGES

- **Academic Foundation:**

- **Faculty of Technical Sciences (FTN), Novi Sad** – *Software and Information Technologies*.

- **Graduation project - SaleTrackMate:**

- **SaaS Invoice Management:** Architected a high-performance system for the full invoice lifecycle, designed to meet the demand for electronic document management in the regional market.
 - **Standardization & Interoperability:** Engineered export modules for **UBL 2.1 (XML)** to ensure compatibility with modern e-Invoicing standards and government portals (SEF), alongside automated **PDF generation**.
 - **Business Intelligence:** Developed an **analytics engine** for real-time tracking of goods, services, and company-wide financial metrics.
 - **Granular Access Control:** Implemented a robust multi-tier RBAC system (Unauthenticated, Standard, Admin) to ensure data isolation and security.

- **eGovernment Ecosystem (Microservices & Golang):**

- Contributed to a distributed system using **Golang** for a manually written **Single Sign-On (SSO)** service.
 - Managed authentication across a microservices mesh integrating judicial and law enforcement modules.

- **Social Media Engine (Twitter Clone & NoSQL Polyglot):**

- Implemented polyglot persistence using **MongoDB** (Users), **Redis** (Caching), **Cassandra** (Feeds), and **Neo4j** (Social Graphs).
 - Utilized **Scrum** methodologies with weekly sprints and retrospectives to ensure iterative delivery.

- **Gym Management System (Spring Boot & MVC):**

- Developed a full-stack portal with e-commerce features using **Spring Boot**, **MySQL**, and **Thymeleaf** (SSR).
 - Engineered custom JDBC-based ORM logic for session booking and loyalty tracking.

- **Linguistic Proficiency:**

- **English (B1/B2):** Professional technical communication; proficient in managing architectural documentation and cross-border technical correspondence.
 - **Serbian:** Native.

INTERESTS & PHILOSOPHY

- **Engineering Mindset:** Prioritizing quality and architectural robustness over pure quantity. I believe in writing code that is auditable, maintainable, and resilient to legacy edge cases.
- **Strategic Thinking:** Enjoying chess as a way to practice pattern recognition and long-term planning.
- **Lifestyle:** Staying active through cycling and walking as a simple way to recharge and stay sharp for long coding sessions.